Day 42 — Optimization & Stability Report

Objective

Improve HR interview AI stability, transcript handling, scoring consistency, and processing reliability without changing the existing scoring weights or thresholds.

Changes Implemented

1. Follow-up stability

- Prevented duplicate follow-up suppression from leaving the engine in an inconsistent eligible state.
- Preserved the existing maximum follow-up limit.
- Added regression coverage for duplicate and maximum-follow-up behavior.

2. Transcript handling

- Normalized repeated whitespace and line breaks in captured interview responses.
- Preserved blank-response behavior after cleanup.
- Added tests for whitespace normalization and blank responses.

3. HR scoring consistency

- Normalized answer whitespace before scoring.
- Added coverage for normalized answers and blank-answer handling.
- Kept existing scoring weights and thresholds unchanged.

Validation Results

Validation	Result
Targeted tests	47 passed
Full regression suite	262 passed
Full regression pytest runtime	22.39 seconds
Measured full-suite elapsed time	24.81 seconds
Warnings	2

Existing warnings

1. Starlette warns that using `httpx` with `starlette.testclient` is deprecated and recommends `httpx2`.
2. `tests/test_ats_system_evaluation.py::test_ats_system_evaluation` returns a dictionary instead of using assertions and returning `None`.

Both warnings were non-fatal. The full regression suite passed.

Evidence and Limitations

- Test results confirm that the implemented stability changes pass the current regression suite.
- No labeled evaluation dataset or controlled before-and-after accuracy benchmark was run for false-positive and false-negative rates.
- No controlled before-and-after performance benchmark was run. The recorded suite runtime alone does not establish a processing-speed improvement.
- Therefore, reductions in classification errors and processing time remain unverified.

Conclusion

Day 42 implementation and regression validation are complete. Follow-up eligibility consistency, transcript whitespace normalization, and HR scoring input cleanup are covered by automated tests. Accuracy and speed improvements require dedicated benchmark evidence before they can be claimed.